

Metal organic framework derived heteroatoms and cyano ($-\text{C}\equiv\text{N}$) group co-decorated porous $\text{g-C}_3\text{N}_4$ nanosheets for improved photocatalytic H_2 evolution and uranium(VI) reduction

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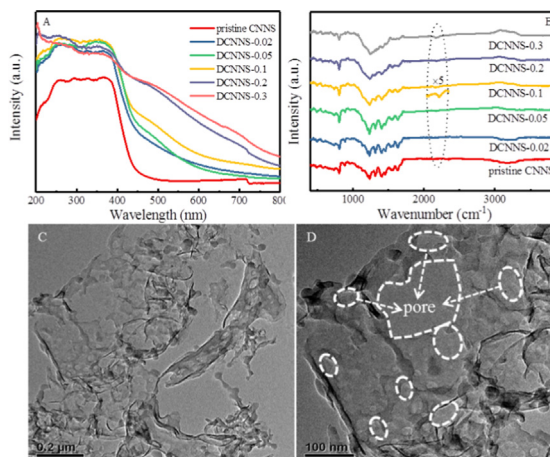
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HIGHLIGHTS

- Heteroatoms and cyano ($-\text{C}\equiv\text{N}$) group co-decorated $\text{g-C}_3\text{N}_4$ nanosheets is synthesized.
- Heteroatoms (Zn + C) doping modulates band gap and prolongs electron lifetime.
- Cyano group broadens light response range and acts as electron-withdrawing group.
- Massive nanopores in nanosheets provide more reaction sites and diffusion channels.
- The optimized photocatalyst displays a maximum H_2 evolution rate of $\sim 484.09 \mu\text{mol/h}$.

GRAPHICAL ABSTRACT

Heteroatoms (Zn + C) and cyano ($-\text{C}\equiv\text{N}$) group co-decorated porous $\text{g-C}_3\text{N}_4$ nanosheets were synthesized, exhibiting enhanced photocatalytic H_2 evolution and U(VI) reduction activity.



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ABSTRACT

The introduction of heteroatoms and functional groups in $\text{g-C}_3\text{N}_4$ generally has great advantages in enhancing the photocatalytic performance. In this work, the heteroatoms (Zn + C) and cyano ($-\text{C}\equiv\text{N}$) group co-decorated porous $\text{g-C}_3\text{N}_4$ nanosheets (DCNNS) photocatalysts were successfully synthesized through direct calcination of the mixed urea and metal-organic frameworks. The optimized DCNNS displayed a maximum H_2 evolution rate of $\sim 484.09 \mu\text{mol/h}$ with a quantum efficiency of $\sim 3.43\%$ at 420 nm, and the photocatalytic U(VI) reduction activity was improved by ~ 6.09 times. The enhanced photocatalytic performance could be ascribed to following benefits: (1) the modified DCNNS shared the two-dimensional layered structure of $\text{g-C}_3\text{N}_4$, and the massive nanopores in the nanosheets provided more reaction sites and diffusion channels for accelerated mass transfer; (2) the formation of cyano group greatly broadened the light response range and also acted as strong electron-withdrawing group for improving the carrier separation rate; (3) heteroatoms doping modulated the band gap, increased the electric conductivity, promoted the carrier separation and transport, and prolonged the electron lifetime

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to enhance photocatalytic performance. This work suggested that the heteroatoms and functional groups co-decoration could significantly improve the performance of g-C₃N₄-based photocatalysts and hold great potential to be further explored for energy and environmental applications.

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1. Introduction

The enormous and increasing consumption of fossil energy has driven humanity to seek new energy sources with lower carbon. Hydrogen energy and nuclear energy are two types of carbon-free energy source and are capable of meeting the growing energy demand. The fast development of nuclear energy surely leads to increasing demands for uranium, which is the key element in nuclear fuels. Therefore, reduction of soluble U(VI) to insoluble U(IV) oxide has been proposed as an important approach to U(VI) extraction from radioactive contamination for further recovery as a fuel [1–3]. Compared to other reduction methods, photocatalytic U(VI) reduction (PUR) has more prospects because it does not produce additional harmful chemicals and is less costly [4,5]. Meanwhile, the photocatalytic H₂ evolution (PHE) was one of the most promising approaches for the production of H₂ from water [6–8]. Therefore, developing low-cost and environmental-friendly photocatalysts is highly desired in the photocatalytic field.

Graphitic carbon nitride (g-C₃N₄), a novel metal-free semiconductor, was a promising visible light photocatalyst because of its relatively narrow band gap (~2.7 eV), well-suited band positions and high stability [9–14]. However, the practical application of pristine g-C₃N₄ in photocatalysis also faced challenges due to its inherent disadvantages [15–17]. Researchers have made significant efforts to enhance the photocatalytic efficiency of g-C₃N₄ resorting to various strategies, such as nanostructure and defect engineering, heteroatoms doping, heterostructure construction or being decorated with co-catalyst [18–29].

Heteroatoms doping was a promising and effective strategy for enhancing the photocatalytic activity of g-C₃N₄ because it could simultaneously regulate the band gap edges, the optical response range and conductivity [30]. For example, Dong et al. proposed that carbon self-doping could reduce the band gap and boost the electrical conductivity of g-C₃N₄ through calcinations with melamine pre-treated with absolute ethanol [31]. Tonda et al. observed that Fe-doped g-C₃N₄ nanosheets exhibited red shift and enhanced visible light response ability [32]. In fact, multi-element doping of g-C₃N₄, such as P + S co-doping, Fe + C co-doping and S + P + O tri-doping have been shown to afford more excellent photocatalytic activity and unusual physiochemical properties in comparison with single element doping [33–35]. However, few studies have been conducted on the metal and nonmetal co-doped g-C₃N₄ for PHE and U(VI) reduction. In particular, most researches on U(VI) reduction were focused on nonmetal modified g-C₃N₄.

Rational modification by functional groups (—NH₂, —OH and —C≡N) was regarded as one of strategies to modulate the activity of g-C₃N₄. [36] These functional groups could improve light absorption ability, modulation band structure, and act as the efficient electron-consuming capture agent [37–40]. For example, Yu et al. grafted hydroxyls on g-C₃N₄ through a hydrothermal process [12]. Such surface hydroxylation allowed g-C₃N₄ to realize local spatial charge separation, accounting for the excellent charge transfer. Zhang and coworkers prepared atomically-thin porous g-C₃N₄ nanosheets with oxygen-containing and amino group through a two-step thermal treatment process [41]. This modification increased the separation and transfer rate of photogenerated electron-hole pairs and provided a large number of reaction sites for surface reactions. However, most functional group modification

methods generally required additional chemical reagents or post-treatment, increasing the complexity of the preparation process. Moreover, the constructed nanosheets structure modulated the migration distance of photogenerated carriers in the vertical-plane direction while the migration distance in-plane direction was still very long, resulting in ineffective suppression of photogenerated electron-hole recombination [42]. The candidate strategy was to cut the g-C₃N₄ nanosheets into interconnected porous nanosheets with abundant in-plane edges, thereby further shortening the migration distance direction of photogenerated carriers in the plane and promoting the penetration of H⁺ into the photocatalyst to generate H₂. Nonetheless, most tailored nanosheets strategies involve drawbacks of rigorous post-processing, operational complexity, and the use of hazardous chemical reagents. It is meaningful and intriguing to develop a simple, green, and no-template new strategy for synthesizing porous g-C₃N₄ nanosheets with interconnected porous.

Based on the above considerations, it is conceivable that g-C₃N₄ with the following three characteristics may enhance its photocatalytic performance: (i) cyano (—C≡N) group acts as a strong electron withdrawing group to accelerate the separation rate of photogenerated carriers, (ii) heteroatoms binary doping alter the band structure, improve the electric conductivity and expand light response range, and (iii) a porous structure with high surface area could provide much more reaction sites. So far, the simultaneous combination cyano (—C≡N) group, heteroatoms binary doping and 2 dimensional (2D) nanosheets with interconnected porous are still scarce.

Metal-organic frameworks (MOFs), constructed from metal ions and organic ligands, were emerging as promising templates and/or precursors to prepare functional nanomaterials via thermolysis [43]. Herein, we present a facile method to synthesize both heteroatoms (Zn + C) and cyano (—C≡N) group co-decorated porous g-C₃N₄ nanosheets through direct calcination of the mixed urea and Zn-MOFs as single precursors. The obtained photocatalysts were characterized, and their photocatalytic H₂ evolution activity for evolution and U(VI) reduction were evaluated. Our work provided a facile strategy to achieve functional groups and heteroatoms co-decorated g-C₃N₄ using MOFs as a single precursor.

2. Experimental section

2.1. Synthesis of pristine g-C₃N₄ nanosheets (CNNS)

Generally, 15 g of urea was heated at 550 °C in an alumina crucible with a cover for 2 h (2.3 °C min^{−1}) in a muffle furnace. The resultant yellow product was collected for further use.

2.2. Synthesis of Zn-MOFs precursor

Typically, zinc nitrate (Zn(NO₃)₂·6H₂O) and hexamethylenetetramine (HMTA) with molar ratio of 5:1 were dissolved in ethanol, respectively. Then, the Zn(NO₃)₂·6H₂O solution was added to the HMTA solution. Afterwards, the precipitates were obtained after stirring for 30 min, centrifugation and washed repeatedly with ethanol, and dried at 80 °C.

2.3. Synthesis of heteroatoms (Zn + C) and cyano ($-\text{C}\equiv\text{N}$) group co-decorated g- C_3N_4 nanosheets (DCNNS)

A series of DCNNS photocatalysts were prepared through direct calcination of urea and Zn-MOFs precursor. Typically, 15 g of urea was ground with Zn-MOFs precursor, and then the solid mixtures were calcined at 550 °C for 4 h ($2.3\text{ }^\circ\text{C min}^{-1}$) in a muffle furnace. The obtained samples were ground for fine powders and marked as DCNNS- x (where $x = 0.02, 0.05, 0.1, 0.2$ and 0.3 , and x corresponds to the mass (g) of Zn-MOFs precursor used).

2.4. Characterization

The morphologies were studied by scanning electron microscope (SEM, JSM-7500) and transmission electron microscope (TEM, JEOL-2100F). The crystal structures were investigated by X-ray diffraction (XRD, Bruker D8 Advanced) equipped with a Cu $\text{K}\alpha$ radiation. The chemical states were learned by X-ray photoelectron spectroscopy (XPS, Thermo Fisher ESCALAB 250Xi). Fourier transform infrared (FTIR) spectra were recorded on a Nicolet-Nexus 670 infrared spectroscopy. Nitrogen adsorption/desorption measurements were conducted with an Autosorb-iQ₂ (Quantachrome Corporation) through nitrogen sorption. The specific surface areas of the samples were calculated using the Brunauer-Emmett-Teller (BET) method. Photoluminescence (PL) data were collected on a Hitachi F4600 Luminescence Spectrometer. UV-vis diffuse reflectance spectra (DRS) were performed using a Shimadzu UV-3600 UV-vis spectrophotometer. Solid electron spin resonance (ESR) signals of samples were investigated by a Bruker model A300 spectrometer at room temperature.

2.5. Photocatalytic H_2 evolution

Typically, photocatalysts (10 mg) were suspended in triethanolamine (TEOA) aqueous solution (10 vol%). $\sim 3\text{ wt}\%$ Pt was loaded on the photocatalysts by an in situ photo-deposition method using H_2PtCl_6 . The suspension was vacuumed thoroughly, and then exposed to a 300 W Xenon lamp ((PLS-SXE, Perfectlight, China)) equipped with a UV cutoff filter ($\lambda \geq 420\text{ nm}$). The amount of evolved H_2 was determined by an online gas chromatography using argon as the carrier gas.

2.6. Photocatalytic U(VI) reduction

Typically, photocatalysts (100 mg) were added in UO_2^{2+} solution (0.12 mM, 200 mL, containing 2.5 mL methanol as sacrificial agent). A 300 W Xe-lamp with a cutoff ($\lambda \geq 420\text{ nm}$) was used as the light source. And then, the solution was stirred in the anaerobic-dark environment for 2 h to achieve adsorption/desorption equilibrium. At certain time intervals, 4 mL of the suspension was analyzed to remove the photocatalyst by centrifugation. The U(VI) concentrations were detected by Shimadzu UV-2600 UV-vis spectrophotometer at 650 nm wavelength.

3. Results and discussion

XRD measurements were used to confirm the phase structures of the samples. It is shown in Fig. 1A that the pristine CNNS exhibited two main characteristic diffraction peaks at $\sim 13.04^\circ$ and $\sim 27.45^\circ$, which can be assigned to the (1 0 0) and (0 0 2) diffraction planes of g- C_3N_4 [44]. The strong peak at $\sim 27.45^\circ$ could be ascribed

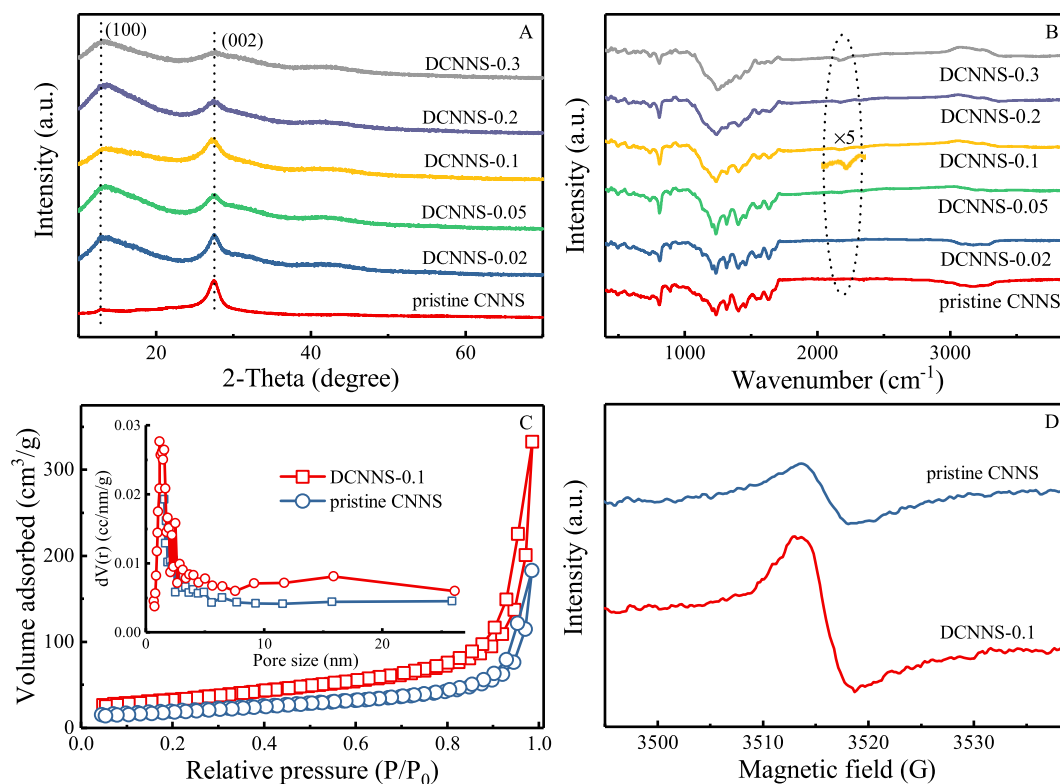


Fig. 1. (A) XRD patterns, (B) FTIR spectra of pristine CNNS and DCNNS- x , (C) N_2 adsorption-desorption isotherms and the BJH pore size distribution curves (inset), and (D) EPR spectra of pristine CNNS and DCNNS-0.1.

to the interlayer stacking of the aromatic system, while the weak peak at $\sim 13.04^\circ$ could be indexed to in-planar-triazine structural packing motif composed of tris-s-triazine units [11]. XRD results suggested well crystalline framework structure of Zn-MOFs (Fig. S1), which was consistent with previously determined MOFs [45]. The characteristic peaks of DCNNS-x gradually became broader and weaker with the increasing amount of Zn-MOFs, indicating the decrease of crystallinity or abundant defects. These differences implied that the original block structure has been transformed into a porous structure. Meanwhile, no characteristic peaks of Zn species were observed, revealing that the incorporation of Zn did not produce a new crystalline phase.

The FTIR spectra of pristine CNNS and DCNNS-x are shown in Fig. 1B. The peaks at $\sim 3000\text{--}3500\text{ cm}^{-1}$ were correlated to —NH and —OH stretching vibrations. The characteristic bands in the range of $\sim 1200\text{--}1650\text{ cm}^{-1}$ and at $\sim 810\text{ cm}^{-1}$ were observed, corresponding to the stretching modes of C—N heterocyclic and the vibrational mode of s-triazine units, respectively [30]. Moreover, a new peak at $\sim 2172\text{ cm}^{-1}$ appeared, which belonged to the cyano ($\text{—C}\equiv\text{N}$) group asymmetric stretching vibration [44,46,47]. The cyano ($\text{—C}\equiv\text{N}$) groups, the electron pairs in which were strongly biased to N, were regarded as strong electron acceptors, thereby contributing to the improvement of the photocatalytic performance.

N_2 adsorption/desorption analysis of pristine CNNS and DCNNS-0.1 was carried out to evaluate their surface area and porosity. As depicted in Figs. 1C and S2, all samples have typical hysteresis loops belonging to type IV isotherms, indicating the multilayer adsorption process of photocatalysts. The surface area of the pristine CNNS, DCNNS-0.02, DCNNS-0.05, DCNNS-0.1, DCNNS-0.2 and DCNNS-0.3 were detected as ~ 68.54 , ~ 73.75 , ~ 85.07 , ~ 115.44 , ~ 84.41 and $\sim 65.23\text{ m}^2/\text{g}$, respectively. Namely, on increasing the Zn-MOF content from 0.02 to 0.1, the surface area of DCNNS increases obviously, and then the surface area decreased with fur-

ther increase of Zn-MOF content from 0.1 to 0.3. A larger surface area could provide more illumination area and more potential reaction sites for H_2 evolution. The insert of Figs. 1C and S2 were the pore size distribution of samples, a similar pore-size (radius) distribution in the region of $\sim 0.5\text{--}30\text{ nm}$ was presented in samples, and a gradually increased peak at $0.5\text{--}10\text{ nm}$ could be clearly observed from pristine CNNS to DCNNS-0.1, manifesting more micropores and mesopores endowed by the Zn-MOF modify. As shown in Table S1, in comparison to pristine CNNS, the average pore sizes of the DCNNS-0.1 have reduced, suggestive of the generation of massive micropores and mesopores. Especially, the DCNNS-0.1 possessed a larger pore volume and pore structure after Zn-MOFs modification, which could facilitate reactants and sacrificial agents across the nanosheets unimpededly during the photocatalytic reactions. The surface area of Zn-MOF was $\sim 3.93\text{ m}^2/\text{g}$, and pore size analyses (the inset in Fig. S3) showed that most of the pores fall into the size range of $\sim 1\text{--}50\text{ nm}$. The pore volume was $\sim 0.0085\text{ cm}^3/\text{g}$.

The unpaired electrons in pristine CNNS and DCNNS-0.1 were studied using EPR technology. In Fig. 1D, the Lorentzian line centered at $g = \sim 2.002$ for pristine CNNS and DCNNS-0.1, were assigned to unpaired electrons in g- C_3N_4 . Compared to the pristine CNNS, the single intensity of DCNNS-0.1 was significantly enhanced, which should be attributed to the redistribution of π -electrons and the introduction of cyano ($\text{—C}\equiv\text{N}$) group defects in DCNNS-0.1.[21] The defects could narrow the band gap to enhance light absorption and improve the photogenerated charge carrier separation.

XPS was carried out to characterize the chemical composition of pristine CNNS and DCNNS-0.1. The survey scan spectra in Fig. 2A revealed the existence of C and N elements in pristine CNNS, while DCNNS-0.1 contain Zn element, besides C and N elements, indicating that the Zn element has been doped into g- C_3N_4 successfully. The C 1s spectra of pristine CNNS (Fig. 2B) displayed three peaks

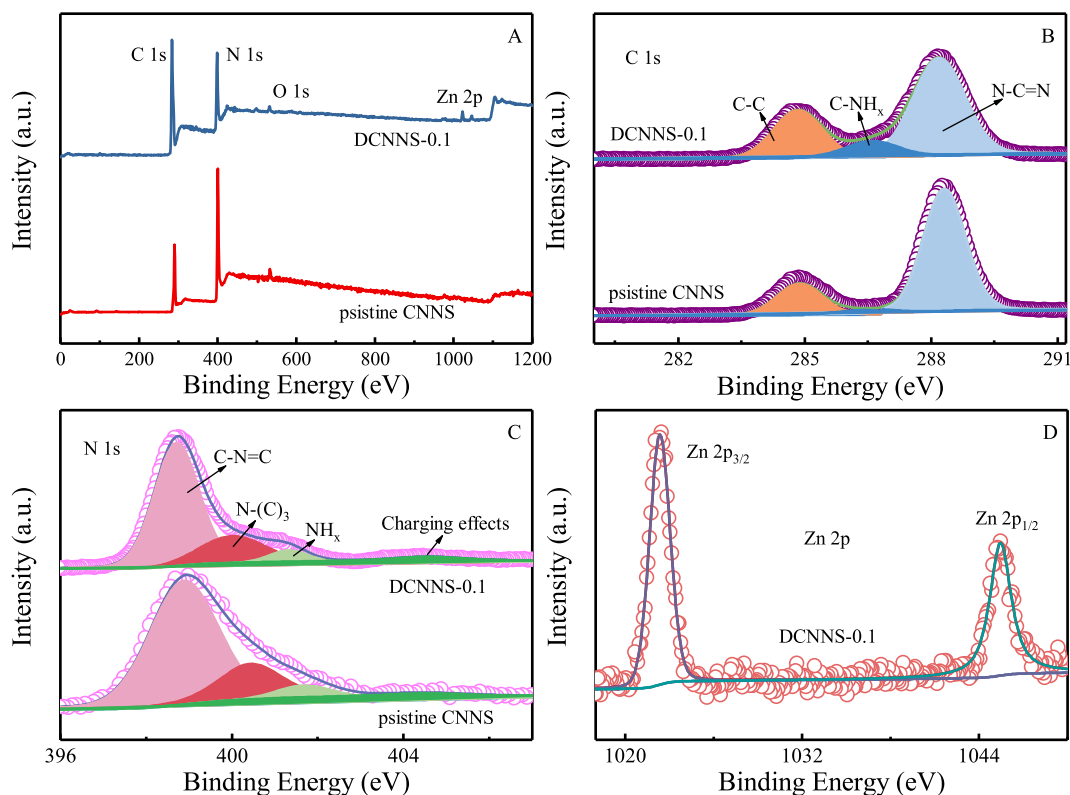


Fig. 2. XPS patterns of pristine CNNS and DCNNS-0.1. (A) Full survey, (B) C 1s, (C) N 1s, and (D) Zn 2p.

at ~ 284.86 , ~ 286.51 and ~ 288.31 eV, associated with graphitic carbon, carbon C-NH_x species and sp^2 hybridized C atom attached to N inside the triazine ring (N-C=N), respectively [8,26]. Notably, the signal of the peak at ~ 286.51 eV of DCNNS-0.1 was intensified compared to pristine CNNS, which could be taken as additional evidence for the generated cyano ($\text{-C}\equiv\text{N}$) group due to the similar binding energy with C-NH_x . Meanwhile, the proportion of peak area at ~ 284.86 eV in DCNNS-0.1 increased compared to that of pristine CNNS ($\sim 10.58\%$ vs. $\sim 2.74\%$), while the proportion of peak area at ~ 288.31 eV in DCNNS-0.1 decreased compared to that of pristine CNNS ($\sim 80.89\%$ vs. $\sim 69.24\%$), clearly demonstrating that the formation of cyano ($\text{-C}\equiv\text{N}$) group might be due to the cleavage of N-C=N heterocycles. As shown in Fig. 2C, the N 1s spectra of pristine CNNS could be resolved into four different peaks at ~ 398.88 , ~ 400.48 , ~ 401.77 and ~ 404.56 eV, which were attributed to the bi-coordinated N (C-N=C), tricoordinated N (N-(C)_3), amino N (C-NH_x) and charging effects in the cyano group, respectively. [15] The binding peaks at ~ 1045.46 and ~ 1022.47 eV, respectively, were designated as $\text{Zn } 2\text{p}_{1/2}$ and $\text{Zn } 2\text{p}_{3/2}$, confirming the presence of Zn element on the DCNNS-0.1 surface, as shown in Fig. 2D [48]. Compared to the pristine CNNS, the atomic ratio of C/N in DCNNS-0.1 increased from ~ 0.64 to ~ 1.13 , confirming the successful incorporation of carbon into $\text{g-C}_3\text{N}_4$ matrix. The carbon doping would optimize the aromatic π conjugated system in $\text{g-C}_3\text{N}_4$ to achieve better photogenerated electron-hole pair separation, and also improved conductivity and photogenerated carrier mobility. The XPS results indicated the occurrence of Zn + C codoping in the $\text{g-C}_3\text{N}_4$ framework.

SEM and TEM were used to observe morphological features of as-prepared samples. The Zn-MOFs precursor presented a 2D nanosheets structure with the size ranging from tens of nanometers to micrometers (Fig. 3A). From the side view, the thickness of nanosheets was ~ 10 nm. The pristine CNNS consisted of thick crimped nanosheets and some aggregates with a size of several micrometers appeared, as shown in Fig. 3B. However, DCNNS-0.1 displayed a curled nanosheets porous worm-like morphology and appeared to be more loosely connected; this scenario was significantly different from the large and flat sheet structure of pristine CNNS sample (Fig. 3C). The pristine CNNS consisted of smooth nanosheets and some loose aggregates with a size of several micrometers appeared (Fig. 3D). It is interesting that the DCNNS-0.1 exhibited strikingly different morphology. Further observation proved the existence of abundant in-plane nanopores of a few nanometers as compared to pristine CNNS, demonstrating the formation of porous carbon nitride sheets (Fig. 3E and F). Therefore, we suspected that the introduction of Zn-MOFs during calcination could etch the surface of pristine CNNS to produce porous $\text{g-C}_3\text{N}_4$ nanosheets. These nanopores should be originated from the burst of soft bubbles which were created by NH_3 and CO_2 gases during the Zn-MOFs condensation [49,50]. No clear lattice fringes were found in HRTEM image (Fig. 3G), indicating the low crystallinity of DCNNS. Additionally, no obvious nanocrystallines could be detected in the SEM and TEM images, further confirming that no other zinc-based species were formed in the DCNNS. This fact was in good agreement with the XRD results. A typical HAAD-STEM images showed uniform dispersion of the C, N and Zn ele-

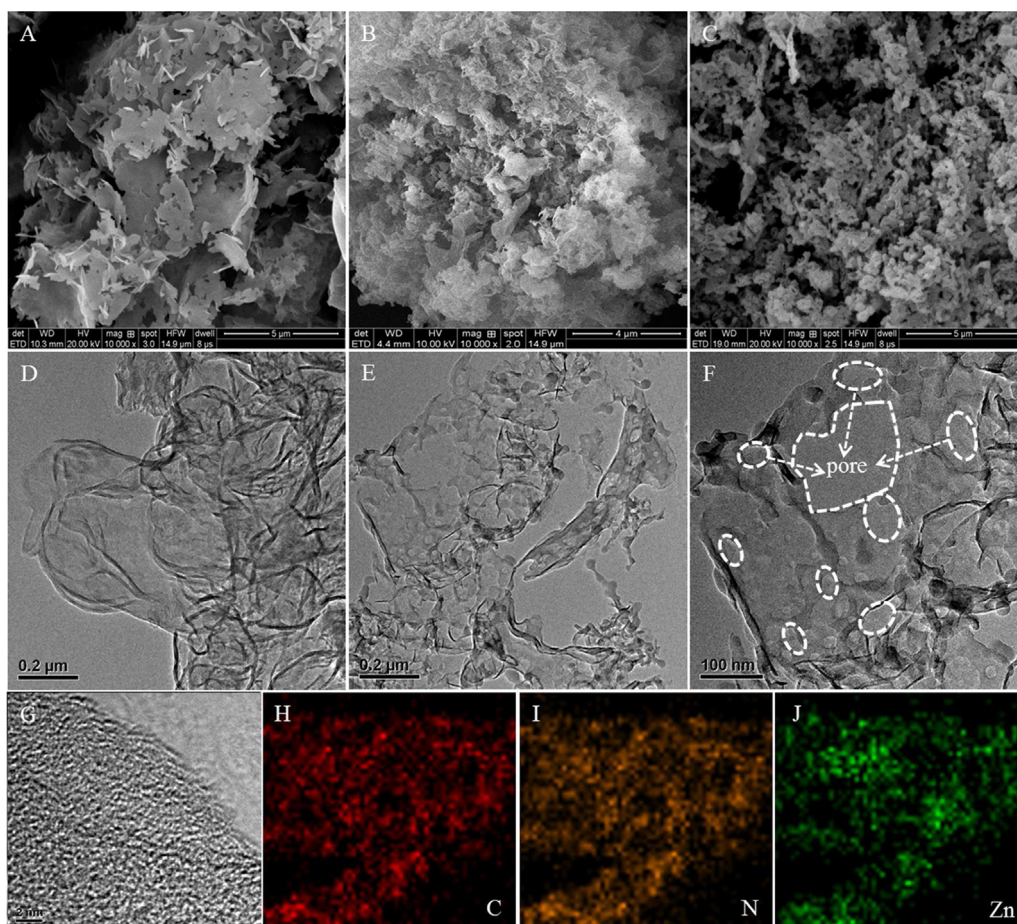


Fig. 3. SEM images of (A) Zn-MOFs, (B) pristine CNNS and (C) DCNNS-0.1, (D) TEM image of pristine CNNS, (E–F) TEM, (G) HRTEM images and (H–J) elements mappings of DCNNS-0.1.

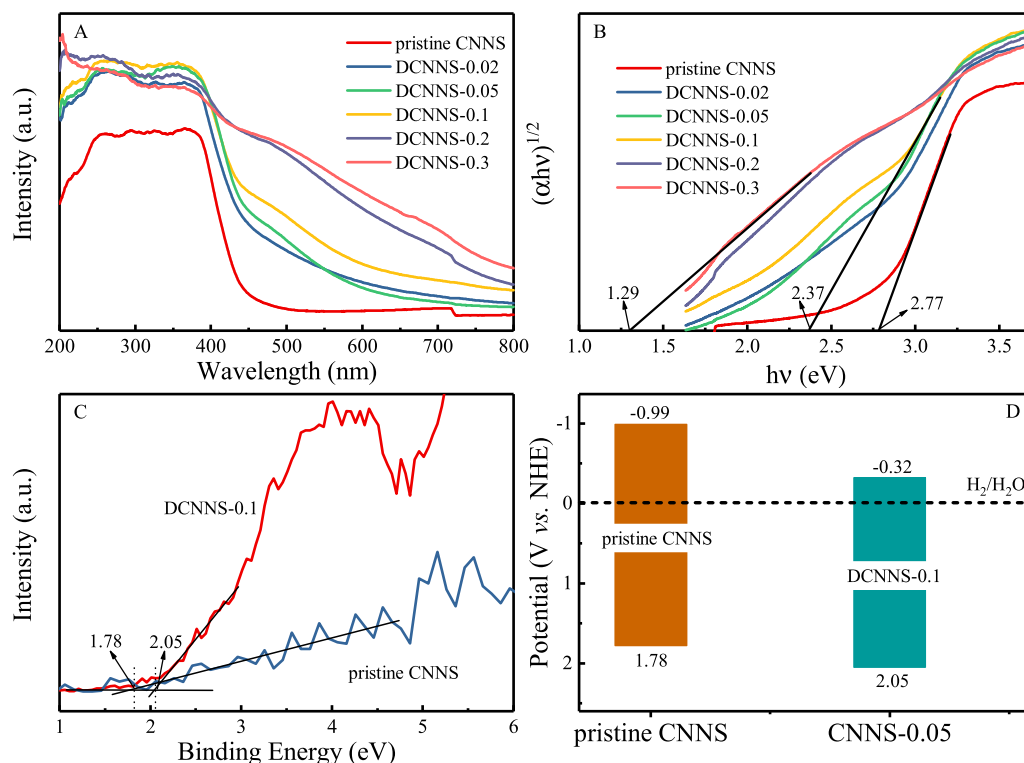


Fig. 4. (A) UV-vis DRS spectra and Tauc plots of $(\alpha h\nu)^{1/2}$ versus $h\nu$ of pristine CNNS and DCNNS-x, (C) VB XPS and (D) band energy diagram of pristine CNNS and DCNNS-0.1.

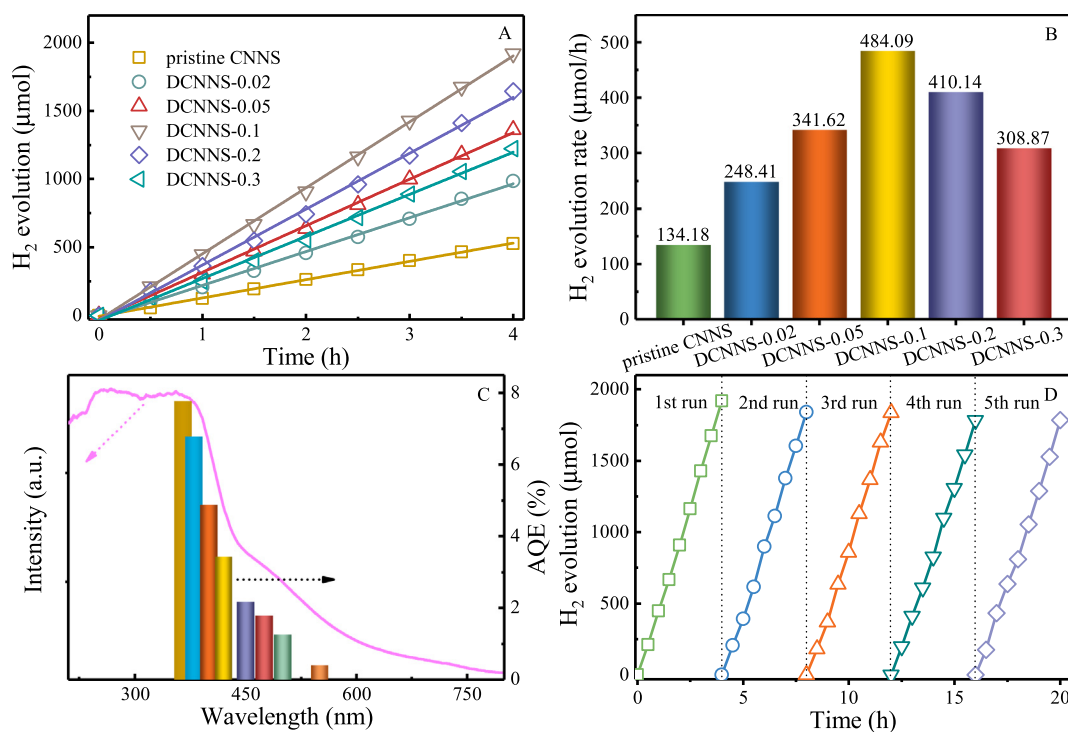


Fig. 5. (A) Time courses of PHE reaction and (B) PHE rates over the pristine CNNS and DCNNS-x, (C) wavelength dependence of AQE on the H_2 evolution in DCNNS-0.1, and (D) cycle H_2 evolution performance of DCNNS-0.1.

ments in the whole nanosheets structure (Fig. 3H–J), further demonstrating the uniform dispersion heteroatoms (Zn + C) doping.

The UV-vis DRS were measured to elucidate the light absorption property of photocatalysts. As shown in Fig. 4A, pristine CNNS

had the absorption edge at ~ 447 nm, while the absorption peaks of DCNNS-x were significantly stronger and broader, along with the red shift of absorption edges from ~ 490 to ~ 960 nm. The increase of both red shift and absorption intensity implied that the samples were more efficient in utilizing sunlight and photogenerated

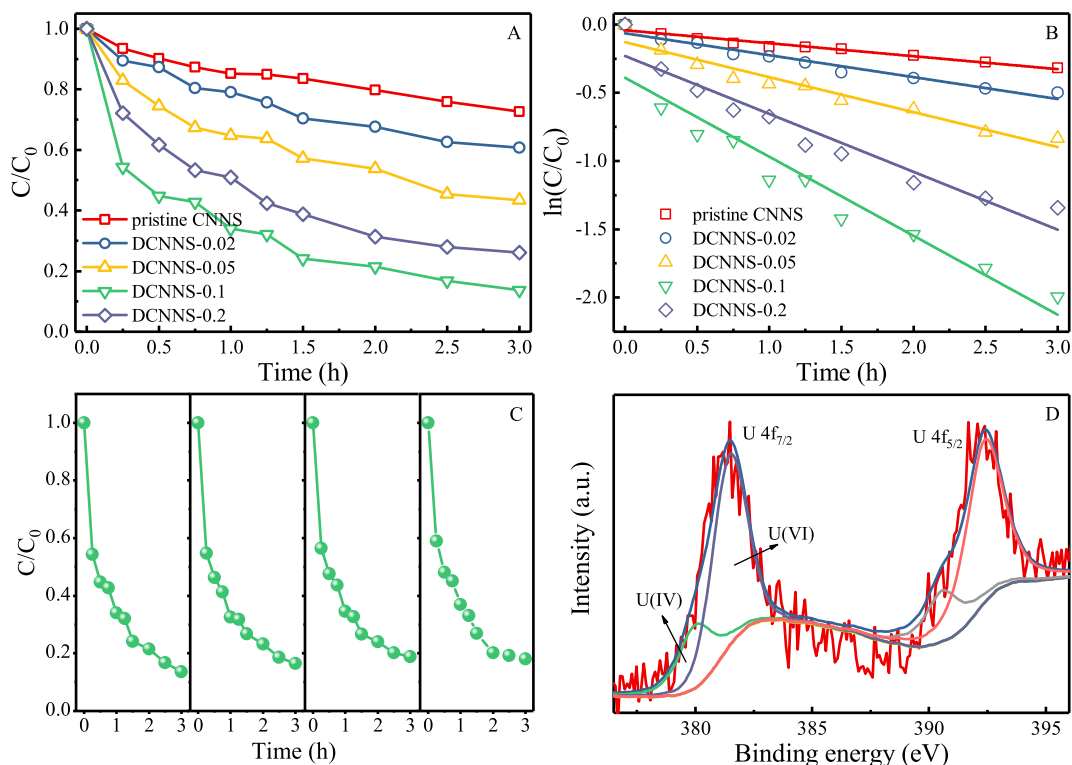


Fig. 6. (A) The PUR activities and (B) reduction kinetics rate constants of pristine CNNS and DCNNS-x, (C) the cycle experiments of DCNNS for PUR and (D) XPS spectra of U(VI) of used DCNNS-0.1.

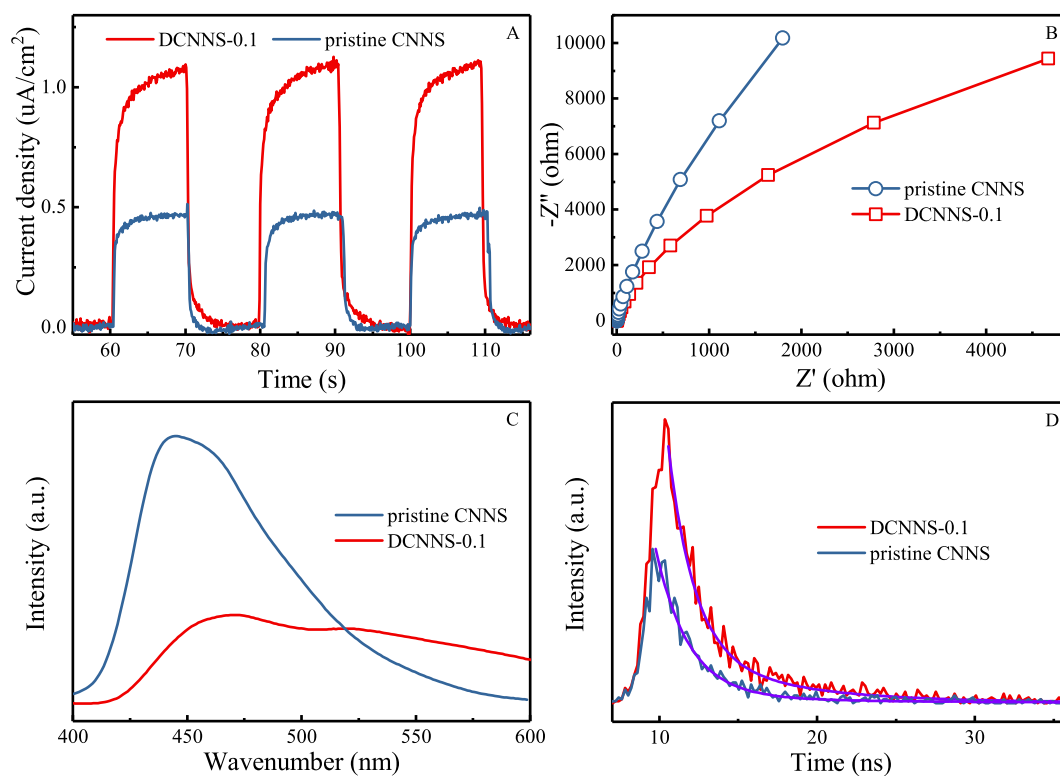


Fig. 7. (A) TPR spectra, (B) EIS curves (c), (C) PL spectra and (D) TRPL spectra of pristine CNNS and DCNNS-0.1.

charge carriers. The band gap (E_g) was determined from $(\alpha h\nu)^{1/2}$ versus $h\nu$ plots, as shown in Fig. 4B. The band gaps for pristine CNNS and DCNNS-0.1 were found to be ~ 2.77 and ~ 2.37 eV, respec-

tively. The DCNNS-0.1 possessed a narrower band gap than pristine CNNS, which might be due to the synergistic effect of heteroatoms (Zn + C) doping and cyano ($-\text{C}\equiv\text{N}$) group. The valence band (VB)

XPS technique was used to obtain the VB position (Fig. 4C). The VB potential of pristine CNNS and DCNNS-0.1 were estimated to be ~ 1.78 and ~ 2.05 V, respectively. According to the band gap energy estimated from UV-vis DRS results, the conduction band (CB) potential of pristine CNNS and DCNNS-0.1 were calculated to be ~ 0.99 and ~ 0.32 eV, respectively. Thus, the CB and VB positions were down-shifted in DCNNS-0.1 in comparison with pristine CNNS, and the detailed band structure diagram is illustrated in Fig. 4D.

The PHE performance of the photocatalysts was comparatively evaluated, and the results are presented in Fig. 5A. The contents of Zn-MOFs significantly impacted the PHE of DCNNS-*x* samples. All of the DCNNS-*x* samples exhibited an enhanced PHE activity compared to pristine CNNS, achieving a maximum for DCNNS-0.1. Doping could reduce the recombination probability of photogenerated charge carriers and prolong photogenerated electrons lifetime by introducing impurity levels into the forbidden band [51]. However, excessive doping could also increase recombination centers for photogenerated electron-hole pairs. The corresponding PHE rates of pristine CNNS and DCNNS-*x* photocatalysts are presented in Fig. 5B. Among them, DCNNS-0.1 exhibited a highest PHE rate of ~ 484.09 $\mu\text{mol/h}$, which was ~ 3.61 times higher than that of pristine CNNS (~ 134.18 $\mu\text{mol/h}$). The apparent quantum efficiency (AQE) for PHE of DCNNS-0.1 was investigated under various monochromatic light irradiation conditions as shown in Fig. 5C. It can be seen that a high AQE of $\sim 3.43\%$ was achieved for DCNNS-0.1 under irradiation at 420 nm. Moreover, the tendency of AQE curve matched well with the UV-vis DRS of DCNNS-0.1, indicating that the PHE reaction was induced by light absorption of $g\text{-C}_3\text{N}_4$. To evaluate the stabilities of DCNNS-0.1, the cycling runs for hydrogen evolution are shown in Fig. 5D. The photocatalytic activity of DCNNS-0.1 did not decrease significantly even after five cycle tests, suggesting its high stability. This method was rather general and could be also used to prepare other heteroatoms and cyano group co-decorated $g\text{-C}_3\text{N}_4$ nanosheets using different metal-based MOF precursor, such as Cu-MOF, Co-MOF, etc. Photocatalytic H_2 production activities over different co-decorated photocatalysts were also evaluated. As shown in Fig. S4, all of co-decorated photocatalysts exhibited considerably enhanced PHE activities. Obviously, DCNNS have the greatest photocatalytic activity. Although the exact reason for this phenomenon was still unclear, it might be attributed to the fact that the Zn-MOF precursor had 2D nanosheets structure, while the Co-MOF and Cu-MOF precursors were bulk structures (Fig. S5). Nanosheet structure precursors were more conducive to the formation of uniform doping.

Furthermore, the PUR under visible light irradiation were carried out to further investigate the underlying environmental application value. As exhibited in Fig. 6A, the PUR rate could be neglected in the absence of photocatalyst and the reduction efficiency of U(VI) over the pristine CNNS was only $\sim 27.4\%$ within 3 h. However, the DCNNS-*x* displayed higher photocatalytic efficiencies than that of pristine CNNS, which implied that the heteroatoms (Zn + C) doping and cyano ($-\text{C}\equiv\text{N}$) group could promoted the separation and migration of photogenerated electron-hole pairs, improved the absorption of visible light, and enhanced the photocatalytic activity. Moreover, the reduction kinetics were evaluated by the pseudo-first-order kinetics equation and shown in Fig. 6B. The kinetics rate constant *k* values were ~ 0.1605 , ~ 0.2559 , ~ 0.5783 and ~ 0.4239 h^{-1} with increasing the amount of Zn-MOFs precursor. The photocatalytic efficiency of the optimum DCNNS-0.1 was ~ 6.09 times higher than that of pristine $g\text{-C}_3\text{N}_4$ (0.0949 h^{-1}). As shown in Fig. 6C, the rate of reduction of U(VI) by DCNNS-0.1 was not significantly deactivated after four cycles, indicating that DCNNS-0.1 had excellent photocatalytic stability. To identify the distribution of the oxidation state of the U element, the high-resolution spectra of U 4f was obtained by XPS

analysis (Fig. 6D). Two peaks at ~ 390.91 and ~ 379.84 eV corresponded to the U $4f_{5/2}$ and U $4f_{7/2}$ of U(IV), respectively, clearly showing the photoreduction of U(VI) [2]. However, XPS results indicated that most of uranium element was still in the form of U(VI), probably because U(IV) was easily reoxidized by photogenerated holes or other free radicals, returning to the initial state U(VI).

The photogenerated charge separation and transfer performance were evaluated by the transient photocurrent response (TPR) and electrochemical impedance spectroscopy (EIS) (Fig. 7). The TPR results showed that the photocurrent density of DCNNS-0.1 was clearly larger than that of pristine CNNS, which demonstrated that the separation and transfer efficiency of photogenerated electro-hole pairs was improved in DCNNS-0.1 (Fig. 7A). As shown in Fig. 7B, DCNNS-0.1 exhibited much smaller EIS Nyquist plot semicircle compared to that of pristine CNNS, indicating that the charge transfer resistance of DCNNS-0.1 was obviously decreased. This fact also implied that the photogenerated charge carriers of DCNNS-0.1 enjoyed higher separation efficiency, which was in accord with the TPR results. Both TPR and EIS results proved the incorporation of heteroatoms (Zn + C) doping and cyano ($-\text{C}\equiv\text{N}$) group could facilitate the charge carrier separation and transfer in photocatalysts. Meanwhile, the porous structure could also reduce the transport distance of the photogenerated charge carrier. PL was further applied to further explore the photogenerated charge carrier recombination process. Fig. 7C showed the room temperature PL spectra of pristine CNNS and DCNNS-0.1 under an excitation of 365 nm. Predictably, the PL intensity of DCNNS-0.1 was dramatically lower than that of pristine CNNS, which indicated that the recombination of electrons and hole pairs could be effectively reduced by introducing the heteroatoms (Zn + C) doping and cyano ($-\text{C}\equiv\text{N}$) group into $g\text{-C}_3\text{N}_4$. Moreover, the PL peak for DCNNS-0.1 showed a blue-shift in contrast to pristine CNNS, which was in accordance with the bandgap blue-shift as shown in the Fig. 4A. Time-resolved PL decay spectroscopy (Fig. 7D) illustrated that the average lifetime of DCNNS-0.1 obviously increased to ~ 4.03 ns compared to pristine CNNS (~ 3.62 ns). The results suggested that a new nonradiative decay pathway might be built, contributing to the prolonged lifetime of photogenerated charge carriers and highly efficient photogenerated electron-hole pairs separation. Overall, the above electrochemical and PL results clearly showed that the heteroatoms (Zn + C) and cyano ($-\text{C}\equiv\text{N}$) group promoted the separation efficiency of electrons-hole pairs, reduced the photocurrent impedance of the material surface, and prolonged the lifetime of photogenerated carriers, thus enhanced reduction effect of U(VI).

Based on above results, the tentative mechanisms of photocatalytic process on the DCNNS are depicted in Fig. 8. The DCNNS

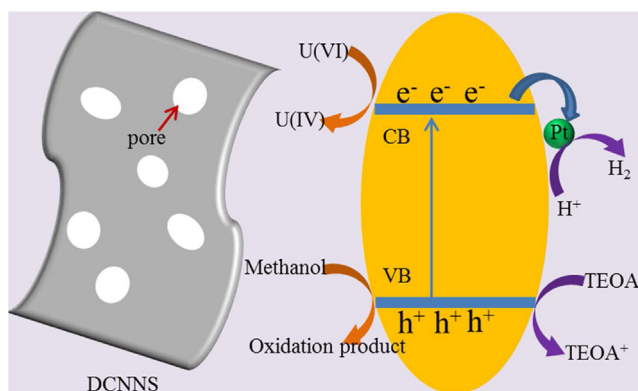


Fig. 8. PHE and PUR mechanisms of DCNNS.

could yield many photogenerated electron-hole pairs under the visible-light illumination. These electrons rapidly injected into the CB and then transferred to the surface. For PHE, these electrons accumulated on the surface of Pt nanoparticles were used for the reduction of H^+ to H_2 . The remained holes were consumed by the sacrificial electron donor of TEOA. For PUR, the photogenerated electrons on the surface of DCNNS could transfer to U(VI), which resulted in the reduction and elimination of U(VI).

4. Conclusions

In summary, we presented a facile strategy to synthesize both cyano ($-C\equiv N$) group and heteroatoms co-decorated porous $g-C_3N_4$ nanosheets through a direct calcination of the mixed urea and Zn-MOFs as single precursors. Their BET surface areas were calculated to be $\sim 73.81 \text{ m}^2/\text{g}$ for pristine CNNS and $\sim 115.44 \text{ m}^2/\text{g}$ for DCNNS-0.1, respectively. A larger surface area could provide more illumination area and more potential reaction sites for PHE and PUR. The ESR results indicated that the presence of a large number of unpaired electrons in DCNNS was due to the formation of cyano ($-C\equiv N$) group. The UV-vis DRS absorption peaks of DCNNS were significantly stronger and broader, and the absorption edges were red shifted from ~ 490 to $\sim 960 \text{ nm}$. The unique dual co-decorated strategy provided faster photogenerated electron-hole separation rates and extended the lifetime of photogenerated carriers, as evidenced by electrochemical analysis and PL/TRPL spectrum. Together with these advantages, the optimum DCNNS-0.1 exhibited a highest hydrogen evolution rate of $\sim 484.09 \mu\text{mol}/\text{h}$, which was ~ 3.61 times higher than that of pristine CNNS. A high AQE of $\sim 3.43\%$ was achieved for DCNNS-0.1 under irradiation at 420 nm . Meanwhile, the DCNNS presented highly efficient PUR activity, was ~ 6.09 times than that of pristine CNNS. Our results suggested that the functional groups and heteroatoms co-decorated as a simple modification technology could significantly improve the $g-C_3N_4$ -based photocatalysts performance and hold a great potential to be further explored for energy and environmental applications.

CRedit authorship contribution statement

Huihui Gao: Data curation, Writing - original draft. **Jinzhong Xu:** Supervision, Validation. **Juntian Zhou:** Software, Visualization, Investigation. **Shouwei Zhang:** Conceptualization, Methodology, Software, Writing - original draft. **Ru Zhou:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2020.02.091>.

References

- [1] X. Wu, S.J. Jiang, S.Q. Song, C.Z. Sun, Constructing effective photocatalytic purification system with P-introduced $g-C_3N_4$ for elimination of UO_2^{2+} , *Appl. Surf. Sci.* 430 (2018) 371–379.
- [2] H. Deng, Z.-J. Li, L. Wang, L.-Y. Yuan, J.-H. Lan, Z.-Y. Chang, Z.-F. Chai, W.-Q. Shi, Nanolayered Ti_3C_2 and $SrTiO_3$ composites for photocatalytic reduction and removal of uranium(VI), *ACS Appl. Nano Mater.* 2 (2019) 2283–2294.
- [3] X.H. Jiang, Q.J. Xing, X.B. Luo, F. Li, J.P. Zou, S.S. Liu, X. Li, X.K. Wang, Simultaneous photoreduction of Uranium(VI) and photooxidation of Arsenic (III) in aqueous solution over $g-C_3N_4/TiO_2$ heterostructured catalysts under simulated sunlight irradiation, *Appl. Catal. B-Environ.* 228 (2018) 29–38.
- [4] C.H. Lu, R.Y. Chen, X. Wu, M.F. Fan, Y.H. Liu, Z.G. Le, S.J. Jiang, S.Q. Song, Boron doped $g-C_3N_4$ with enhanced photocatalytic UO_2^{2+} reduction performance, *Appl. Surf. Sci.* 360 (2016) 1016–1022.
- [5] J. Wang, Y. Wang, W. Wang, Z. Ding, R. Geng, P. Li, D. Pan, J. Liang, H. Qin, Q. Fan, Tunable mesoporous $g-C_3N_4$ nanosheets as a metal-free catalyst for enhanced visible-light-driven photocatalytic reduction of U(VI), *Chem. Eng. J.* 123193 (2019).
- [6] X. Fan, L. Zhang, R. Cheng, M. Wang, M. Li, Y. Zhou, J. Shi, Construction of graphitic C_3N_4 -based intramolecular donor-acceptor conjugated copolymers for photocatalytic hydrogen evolution, *ACS Catal.* 5 (2015) 5008–5015.
- [7] B. Antil, L. Kumar, K.P. Reddy, C.S. Gopinath, S. Deka, Direct thermal polymerization approach to N-rich holey carbon nitride nanosheets and their promising photocatalytic H_2 evolution and charge-storage activities, *ACS Sustain. Chem. Eng.* 7 (2019) 9428–9438.
- [8] Y.Y. Wang, S. Zhao, Y.W. Zhang, J.S. Fang, W.X. Chen, S.H. Yuan, Y.M. Zhou, Facile synthesis of self-assembled $g-C_3N_4$ with abundant nitrogen defects for photocatalytic hydrogen evolution, *ACS Sustain. Chem. Eng.* 6 (2018) 10200–10210.
- [9] W.-J. Ong, L.-L. Tan, Y.H. Ng, S.-T. Yong, S.-P. Chai, Graphitic carbon nitride ($g-C_3N_4$)-based photocatalysts for artificial photosynthesis and environmental remediation: are we a step closer to achieving sustainability?, *Chem. Rev.* 116 (2016) 7159–7329.
- [10] D.D. Chen, J.G. Liu, Z.Z. Jia, J.Z. Fang, F. Yang, Y.M. Tang, K. Wu, Z. Liu, Z.Q. Fang, Efficient visible-light-driven hydrogen evolution and Cr(VI) reduction over porous P and Mo co-doped $g-C_3N_4$ with feeble N vacancies photocatalyst, *J. Hazard. Mater.* 361 (2019) 294–304.
- [11] M. Wu, Y.S. Gong, T. Nie, J. Zhang, R. Wang, H.W. Wang, B.B. He, Template-free synthesis of nanocage-like $g-C_3N_4$ with high surface area and nitrogen defects for enhanced photocatalytic H₂ activity, *J. Mater. Chem. A* 7 (2019) 5324–5332.
- [12] S.X. Yu, J.Y. Li, Y.H. Zhang, M. Li, F. Dong, T.R. Zhang, H.W. Huang, Local spatial charge separation and proton activation induced by surface hydroxylation promoting photocatalytic hydrogen evolution of polymeric carbon nitride, *Nano Energy* 50 (2018) 383–392.
- [13] Z.F. Chen, S.C. Lu, Q.L. Wu, F. He, N.Q. Zhao, C.N.A. He, C.S. Shi, Salt-assisted synthesis of 3D open porous $g-C_3N_4$ decorated with cyano groups for photocatalytic hydrogen evolution, *Nanoscale* 10 (2018) 3008–3013.
- [14] J. Liu, Y. Liu, N.Y. Liu, Y.Z. Han, X. Zhang, H. Huang, Y. Lifshitz, S.T. Lee, J. Zhong, Z.H. Kang, Metal-free efficient photocatalyst for stable visible water splitting via a two-electron pathway, *Science* 347 (2015) 970–974.
- [15] H. Yu, R. Shi, Y. Zhao, T. Bian, Y. Zhao, C. Zhou, G.I.N. Waterhouse, L.-Z. Wu, C.-H. Tung, T. Zhang, Alkali-assisted synthesis of nitrogen deficient graphitic carbon nitride with tunable band structures for efficient visible-light-driven hydrogen evolution, *Adv. Mater.* 29 (2017) 1605148.
- [16] N. Tian, Y. Zhang, X. Li, K. Xiao, X. Du, F. Dong, G.I.N. Waterhouse, T. Zhang, H. Huang, Precursor-reforming protocol to 3D mesoporous $g-C_3N_4$ established by ultrathin self-doped nanosheets for superior hydrogen evolution, *Nano Energy* 38 (2017) 72–81.
- [17] C. Liu, H. Huang, L. Ye, S. Yu, N. Tian, X. Du, T. Zhang, Y. Zhang, Intermediate-mediated strategy to horn-like hollow mesoporous ultrathin $g-C_3N_4$ tube with spatial anisotropic charge separation for superior photocatalytic H_2 evolution, *Nano Energy* 41 (2017) 738–748.
- [18] Z. Lin, X. Wang, Nanostructure engineering and doping of conjugated carbon nitride semiconductors for hydrogen photosynthesis, *Angew. Chem. Int. Edit.* 52 (2013) 1735–1738.
- [19] Z. Tong, D. Yang, Z. Li, Y. Nan, F. Ding, Y. Shen, Z. Jiang, Thylakoid-inspired multishell $g-C_3N_4$ nanocapsules with enhanced visible-light harvesting and electron transfer properties for high-efficiency photocatalysis, *ACS Nano* 11 (2017) 1103–1112.
- [20] Z. Hong, B. Shen, Y. Chen, B. Lin, B. Gao, Enhancement of photocatalytic H_2 evolution over nitrogen-deficient graphitic carbon nitride, *J. Mater. Chem. A* 1 (2013) 11754–11761.
- [21] Q. Liang, Z. Li, Z.-H. Huang, F. Kang, Q.-H. Yang, Holey graphitic carbon nitride nanosheets with carbon vacancies for highly improved photocatalytic hydrogen production, *Adv. Funct. Mater.* 25 (2015) 6885–6892.
- [22] G. Liu, P. Niu, C. Sun, S.C. Smith, Z. Chen, G.Q. Lu, H.-M. Cheng, Unique electronic structure induced high photoreactivity of sulfur-doped graphitic C_3N_4 , *J. Am. Chem. Soc.* 132 (2010) 11642–11648.
- [23] J. Fu, J. Yu, C. Jiang, B. Cheng, $g-C_3N_4$ -based heterostructured photocatalysts, *Adv. Energy Mater.* 8 (2018) 1701503.
- [24] W. Gu, F. Lu, C. Wang, S. Kuga, L. Wu, Y. Huang, M. Wu, Face-to-face interfacial assembly of ultrathin $g-C_3N_4$ and anatase TiO_2 nanosheets for enhanced solar photocatalytic activity, *ACS Appl. Mater. Interfaces* 9 (2017) 28674–28684.

- [25] M. Zhu, S. Kim, L. Mao, M. Fujitsuka, J. Zhang, X. Wang, T. Majima, Metal-free photocatalyst for H_2 evolution in visible to near-infrared region: black phosphorus/graphitic carbon nitride, *J. Am. Chem. Soc.* 139 (2017) 13234–13242.
- [26] L. Ma, H. Fan, K. Fu, S. Lei, Q. Hu, H. Huang, G. He, Protonation of graphitic carbon nitride ($g-C_3N_4$) for an electrostatically self-assembling carbon@ $g-C_3N_4$ core-shell nanostructure toward high hydrogen evolution, *ACS Sustain. Chem. Eng.* 5 (2017) 7093–7103.
- [27] L. Yao, D. Wei, Y. Ni, D. Yan, C. Hu, Surface localization of CdZnS quantum dots onto 2D $g-C_3N_4$ ultrathin microribbons: highly efficient visible light-induced H_2 -generation, *Nano Energy* 26 (2016) 248–256.
- [28] Y. Li, Y. Sun, W. Ho, Y. Zhang, H. Huang, Q. Cai, F. Dong, Highly enhanced visible-light photocatalytic NO_x purification and conversion pathway on self-structurally modified $g-C_3N_4$ nanosheets, *Sci. Bull.* 63 (2018) 609–620.
- [29] H. Gao, R. Cao, S. Zhang, H. Yang, X. Xu, Three-dimensional hierarchical $g-C_3N_4$ architectures assembled by ultrathin self-doped nanosheets: extremely facile hexamethylenetetramine activation and superior photocatalytic hydrogen evolution, *ACS Appl. Mater. Interfaces* 11 (2019) 2050–2059.
- [30] J. Huang, D. Li, R. Li, Q. Zhang, T. Chen, H. Liu, Y. Liu, W. Lv, G. Liu, An efficient metal-free phosphorus and oxygen co-doped $g-C_3N_4$ photocatalyst with enhanced visible light photocatalytic activity for the degradation of fluoroquinolone antibiotics, *Chem. Eng. J.* 374 (2019) 242–253.
- [31] G. Dong, K. Zhao, L. Zhang, Carbon self-doping induced high electronic conductivity and photoreactivity of $g-C_3N_4$, *Chem. Commun.* 48 (2012) 6178–6180.
- [32] S. Tonda, S. Kumar, S. Kandula, V. Shanker, Fe-doped and -mediated graphitic carbon nitride nanosheets for enhanced photocatalytic performance under natural sunlight, *J. Mater. Chem. A* 2 (2014) 6772–6780.
- [33] C. Hu, W.-Z. Hung, M.-S. Wang, P.-J. Lu, Phosphorus and sulfur codoped $g-C_3N_4$ as an efficient metal-free photocatalyst, *Carbon* 127 (2018) 374–383.
- [34] S. Zhang, J. Li, M. Zeng, J. Li, J. Xu, X. Wang, Bandgap engineering and mechanism study of nonmetal and metal ion codoped carbon nitride: C+Fe as an example, *Chem. -Eur. J.* 20 (2014) 9805–9812.
- [35] Q. Liu, J. Shen, X. Yu, X. Yang, W. Liu, J. Yang, H. Tang, H. Xu, H. Li, Y. Li, J. Xu, Unveiling the origin of boosted photocatalytic hydrogen evolution in simultaneously (S, P, O)-Codoped and exfoliated ultrathin $g-C_3N_4$ nanosheets, *Appl. Catal. B-Environ.* 248 (2019) 84–94.
- [36] S. Zhang, S. Song, P.C. Gu, R. Ma, D.L. Wei, G.X. Zhao, T. Wen, R. Jehan, B.W. Hu, X.K. Wang, Visible-light-driven activation of persulfate over cyano and hydroxyl group co-modified mesoporous $g-C_3N_4$ for boosting bisphenol A degradation, *J. Mater. Chem. A* 7 (2019) 5552–5560.
- [37] X.L. Liu, P. Wang, H.S. Zhai, Q.Q. Zhang, B.B. Huang, Z.Y. Wang, Y.Y. Liu, Y. Dai, X.Y. Qin, X.Y. Zhang, Synthesis of synergetic phosphorus and cyano groups (–C equivalent to N) modified $g-C_3N_4$ for enhanced photocatalytic H_2 production and CO_2 reduction under visible light irradiation, *Appl. Catal. B-Environ.* 232 (2018) 521–530.
- [38] H. Tan, X.M. Gu, P. Kong, Z. Lian, B. Li, Z.F. Zheng, Cyano group modified carbon nitride with enhanced photoactivity for selective oxidation of benzylamine, *Appl. Catal. B-Environ.* 242 (2019) 67–75.
- [39] J. Yang, Y. Liang, K. Li, G. Yang, K. Wang, R. Xu, X. Xie, One-step synthesis of novel K^+ and cyano groups decorated triazine/heptazine-based $g-C_3N_4$ tubular homojunctions for boosting photocatalytic H_2 evolution, *Appl. Catal. B-Environ.* 262 (2020) 118252.
- [40] W. Wang, H. Zhang, S. Zhang, Y. Liu, G. Wang, C. Sun, H. Zhao, Potassium-ion-assisted regeneration of active cyano groups in carbon nitride nanoribbons: visible-light-driven photocatalytic nitrogen reduction, *Angew. Chem. Int. Ed.* 58 (2019) 16644–16650.
- [41] N.N. Meng, J. Ren, Y. Liu, Y. Huang, T. Petit, B. Zhang, Engineering oxygen-containing and amino groups into two-dimensional atomically-thin porous polymeric carbon nitrogen for enhanced photocatalytic hydrogen production, *Energ. Environ. Sci.* 11 (2018) 566–571.
- [42] B. Lin, H. An, X.Q. Yan, T.X. Zhang, J.J. Wei, G.D. Yang, Fish-scale structured $g-C_3N_4$ nanosheet with unusual spatial electron transfer property for high-efficiency photocatalytic hydrogen evolution, *Appl. Catal. B-Environ.* 210 (2017) 173–183.
- [43] Y. Gong, X. Zhao, H. Zhang, B. Yang, K. Xiao, T. Guo, J.J. Zhang, H.X. Shao, Y.B. Wang, G. Yu, MOF-derived nitrogen doped carbon modified $g-C_3N_4$ heterostructure composite with enhanced photocatalytic activity for bisphenol A degradation with peroxydisulfate under visible light irradiation, *Appl. Catal. B-Environ.* 233 (2018) 35–45.
- [44] J. Yang, Y. Liang, K. Li, G. Yang, K. Wang, R. Xu, X. Xie, Cyano and potassium-rich $g-C_3N_4$ hollow tubes for efficient visible-light-driven hydrogen evolution, *Catal. Sci. Technol.* 9 (2019) 3342–3346.
- [45] S.T. Liu, J.S. Zhou, H.H. Song, 2D Zn-hexamine coordination frameworks and their derived N-rich porous carbon nanosheets for ultrafast sodium storage, *Adv. Energy Mater.* 8 (2018).
- [46] D. Zhang, X.H. Han, T. Dong, X.W. Guo, C.S. Song, Z.K. Zhao, Promoting effect of cyano groups attached on $g-C_3N_4$ nanosheets towards molecular oxygen activation for visible light-driven aerobic coupling of amines to imines, *J. Catal.* 366 (2018) 237–244.
- [47] Y. Xue, Y. Guo, Z. Liang, H. Cui, J. Tian, Porous $g-C_3N_4$ with nitrogen defects and cyano groups for excellent photocatalytic nitrogen fixation without co-catalysts, *J. Colloid Interface Sci.* 556 (2019) 206–213.
- [48] Q.H. Yang, C.C. Yang, C.H. Lin, H.L. Jiang, Metal-organic-framework-derived hollow N-doped porous carbon with ultrahigh concentrations of single Zn atoms for efficient carbon dioxide conversion, *Angew. Chem. Int. Edit.* 58 (2019) 3511–3515.
- [49] S.T. Liu, J.S. Zhou, H.H. Song, Tailoring highly N-doped carbon materials from hexamine-based MOFs: superior performance and new insight into the roles of N configurations in Na-ion storage, *Small* 14 (2018).
- [50] D.D. Sun, S.T. Liu, G.J. Zhang, J.S. Zhou, $NiTe_2/N$ -doped graphitic carbon nanosheets derived from Ni-hexamine coordination frameworks for Na-ion storage, *Chem. Eng. J.* 359 (2019) 1659–1667.
- [51] L.B. Jiang, X.Z. Yuan, Y. Pan, J. Liang, G.M. Zeng, Z.B. Wu, H. Wang, Doping of graphitic carbon nitride for photocatalysis: a review, *Appl. Catal. B-Environ.* 217 (2017) 388–406.